1 Planning (15 marks)

Defining the problem (3 marks)

P *V* is the independent variable, or vary *V* **and** *f* is the dependent variable, or measure *f*.
Or *f* is the independent variable, or vary *f* **and** *V* is the dependent variable, or measure *V*. [1]

P Change *f* (allow *V*) until the mass leaves/gap between plate. [1]

P Keep the position of the mass constant. (Do not allow keep mass constant.) [1]

Methods of data collection (5 marks)

M Labelled diagram showing signal generator/a.c. supply connected to vibrator with two wires with mass on plate. At least two labels needed. [1]

M Voltmeter/c.r.o. connected in parallel with vibrator in a workable circuit. [1]

M Measure *f* or *T* from signal generator/c.r.o. (Allow detailed use of motion sensor/stroboscope.) [1]

M Detail regarding mass leaving the plate: listen to noise, look for gap. [1]

M Repeat each experiment for the same value of *V* (allow *f* if consistent with above) and average. [1]

Method of analysis (2 marks)

Plot a graph of:

A f^2 against $1/V$ | $1/V$ against f^2 | f against $1/\sqrt{V}$ | $1/\sqrt{V}$ against f | $\lg V$ against $\lg f$ | $\lg f$ against $\lg V$

or

V against $1/f^2$ | $1/f^2$ against V | $\sqrt{V}$ against $1/f$ | $1/f$ against $\sqrt{V}$ [1]

A $k =$ gradient $\times \pi^2$ | $k = \dfrac{\pi^2}{\text{gradient}}$ | $k =$ gradient$^2 \times \pi^2$ | $k = \dfrac{\pi^2}{\text{gradient}^2}$ | $k = \pi^2 \times 10^c$ | $k = \pi^2 \times 10^{2c}$ [1]

Safety considerations (1 mark)

S Precaution linked to mass leaving vibrating plate, e.g. use safety screen/goggles/sand tray. [1]

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Additional detail (4 marks)

D Relevant points might include [4]
1 Wait for vibrator to oscillate evenly
2 Method to determine period of oscillation from c.r.o., i.e. one time period × time-base
3 Method to determine *f* from c.r.o. having determined *T*, i.e. $f = 1 / T$
4 Method to determine *V* from c.r.o, i.e. amplitude (height) × *y*-gain
5 Relationship is valid if the graph is a straight line passing through the origin
[For lg – lg graph the gradient must be correct (–2 or –0.5)]
6 Determine *f* (allow *V* if consistent with above) by increasing and decreasing *V* or *f*
7 Clean surfaces of metal plate/small mass
8 Spirit level to keep plate horizontal/eye level to look for gap

Do not allow vague computer methods.

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